

# PERMUTATION FEATURE IMPORTANCE

$$\mathcal{R}_{\text{emp}}(\hat{f}, \tilde{\mathcal{D}}_{(k)}^S) - \mathcal{R}_{\text{emp}}(\hat{f}, \mathcal{D})$$

| $i$ | $\mathbf{x}_S$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ | $\mathbf{x}_1$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ | $\Delta L$ |
|-----|----------------|----------------|----------------|----------------|----------------|----------------|------------|
| 1   | 2              | 4              | 7              | 1              | 4              | 7              | 0.65       |
|     | 1              | 5              | 8              | 2              | 5              | 8              | 0.15       |
|     | 3              | 6              | 9              | 3              | 6              | 9              | 0          |
| m   | 3              | 4              | 7              | 1              | 4              | 7              | 0.85       |
|     | 2              | 5              | 8              | 2              | 5              | 8              | 0          |
|     | 1              | 6              | 9              | 3              | 6              | 9              | 0.35       |

$= 0.267$   
 $\widehat{PFI}_S = \frac{1}{2} (0.267 + 0.4)$   
 $= 0.4$

## 3. Aggregation:

- Compute the loss for each observation in both data sets
- Take the difference of both losses  $\Delta L$  for each observation
- Average this change in loss across all observations
- Repeat perturbation and average over multiple repetitions